

Week 5 & 6 Assignment - Kia Thefts

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```
library(readxl)
vice <- read_excel("Motherboard VICE News Kia Hyundai Theft Data.xlsx")
head(vice, 10)
```

```
## # A tibble: 10 x 211
##   ...1      Denver ...3 ...4 'El Paso' ...6 ...7 Portland ...9
##   <dtm>      <chr>  <chr> <chr> <chr>      <chr> <chr> <chr>  <chr>
## 1 NA      Kia/Hyu~ All   Perc~ Kia/Hyun~ All   Perc~ Kia/Hyu~ All
## 2 2019-12-01 00:00:00 48.0    615.0 0.07~ 13.0    103.0 0.12~ 13.0    592.0
## 3 2020-01-01 00:00:00 21.0    519.0 0.04~ 9.0     95.0  0.09~ 12.0    559.0
## 4 2020-02-01 00:00:00 28.0    402.0 0.06~ 5.0     64.0  0.07~ 10.0    498.0
## 5 2020-03-01 00:00:00 35.0    508.0 0.06~ 6.0     75.0  0.08  4.0     465.0
## 6 2020-04-01 00:00:00 39.0    616.0 0.06~ 1.0     57.0  0.01~ 7.0     488.0
## 7 2020-05-01 00:00:00 37.0    692.0 0.05~ 1.0     62.0  0.01~ 7.0     387.0
## 8 2020-06-01 00:00:00 57.0    747.0 0.07~ 8.0     77.0  0.10~ 11.0    423.0
## 9 2020-07-01 00:00:00 57.0    786.0 0.07~ 5.0     62.0  0.08~ 11.0    528.0
## 10 2020-08-01 00:00:00 85.0    840.0 0.10~ 4.0     83.0  0.04~ 8.0     629.0
## # i 202 more variables: ...10 <chr>, Atlanta <chr>, ...12 <chr>, ...13 <chr>,
## #   Chicago <chr>, ...15 <chr>, ...16 <chr>, 'Virginia Beach' <chr>,
## #   ...18 <chr>, ...19 <chr>, Louisville <chr>, ...21 <chr>, ...22 <chr>,
## #   'San Diego' <chr>, ...24 <chr>, ...25 <chr>, Sacramento <chr>, ...27 <chr>,
## #   ...28 <chr>, Dallas <chr>, ...30 <chr>, ...31 <chr>, 'Fort Worth' <chr>,
## #   ...33 <chr>, ...34 <chr>, Bakersfield <chr>, ...36 <chr>, ...37 <chr>,
## #   'Omaha, NE' <chr>, ...39 <chr>, ...40 <chr>, 'Lubbock, TX' <chr>, ...
```

```
library(tidyverse)
colnames(vice)
```

```
##   [1] "...1"      "Denver"
##   [3] "...3"      "...4"
##   [5] "El Paso"   "...6"
##   [7] "...7"      "Portland"
##   [9] "...9"      "...10"
##  [11] "Atlanta"   "...12"
##  [13] "...13"     "Chicago"
##  [15] "...15"     "...16"
##  [17] "Virginia Beach" "...18"
##  [19] "...19"     "Louisville"
##  [21] "...21"     "...22"
##  [23] "San Diego" "...24"
##  [25] "...25"     "Sacramento"
```

## [27]	"...27"	"...28"
## [29]	"Dallas"	"...30"
## [31]	"...31"	"Fort Worth"
## [33]	"...33"	"...34"
## [35]	"Bakersfield"	"...36"
## [37]	"...37"	"Omaha, NE"
## [39]	"...39"	"...40"
## [41]	"Lubbock, TX"	"...42"
## [43]	"...43"	"Fremont, CA"
## [45]	"...45"	"...46"
## [47]	"San Antonio"	"...48"
## [49]	"...49"	"Irving, Texas"
## [51]	"...51"	"...52"
## [53]	"Madison, WI"	"...54"
## [55]	"...55"	"Frisco, TX"
## [57]	"...57"	"...58"
## [59]	"Lincoln, NE"	"...60"
## [61]	"...61"	"Wichita, KS"
## [63]	"...63"	"...64"
## [65]	"Plano, TX"	"...66"
## [67]	"...67"	"Akron, OH"
## [69]	"...69"	"...70"
## [71]	"Buffalo, NY"	"...72"
## [73]	"...73"	"Chula Vista, CA"
## [75]	"...75"	"...76"
## [77]	"Reno, NV"	"...78"
## [79]	"...79"	"Oxnard, CA"
## [81]	"...81"	"...82"
## [83]	"San Francisco"	"...84"
## [85]	"...85"	"McKinney, TX"
## [87]	"...87"	"...88"
## [89]	"Arlington, TX"	"...90"
## [91]	"...91"	"Garland, TX"
## [93]	"...93"	"...94"
## [95]	"Riverside County, CA"	"...96"
## [97]	"...97"	"Stockton, CA"
## [99]	"...99"	"...100"
## [101]	"Corpus Cristi, TX"	"...102"
## [103]	"...103"	"Cincinatti"
## [105]	"...105"	"...106"
## [107]	"Montgomery County, MD"	"...108"
## [109]	"...109"	"Henderson, NV"
## [111]	"...111"	"...112"
## [113]	"San Bernardino, CA"	"...114"
## [115]	"...115"	"Newport News, VA"
## [117]	"...117"	"...118"
## [119]	"Amarillo, TX"	"...120"
## [121]	"...121"	"New Haven, CT"
## [123]	"...123"	"...124"
## [125]	"Washington, D.C."	"...126"
## [127]	"...127"	"Boise, ID"
## [129]	"...129"	"...130"
## [131]	"Austin, TX"	"...132"
## [133]	"...133"	"Chandler, AZ"

```
## [135] "...135"                "...136"
## [137] "Modesto, CA"           "...138"
## [139] "...139"                "San Jose, CA"
## [141] "...141"                "...142"
## [143] "Aurora, IL"            "...144"
## [145] "...145"                "Vancouver, WA"
## [147] "...147"                "...148"
## [149] "Raleigh, NC"           "...150"
## [151] "...151"                "St. Petersburg, FL"
## [153] "...153"                "...154"
## [155] "Milwaukee, WI"         "...156"
## [157] "...157"                "Los Angeles, CA"
## [159] "...159"                "...160"
## [161] "Glendale, AZ"          "...162"
## [163] "...163"                "Orlando, FL"
## [165] "...165"                "...166"
## [167] "Eugene, OR"            "...168"
## [169] "...169"                "Rochester, NY"
## [171] "...171"                "...172"
## [173] "Durham, NC"            "...174"
## [175] "...175"                "Norfolk, VA"
## [177] "...177"                "...178"
## [179] "Pittsburgh, PA"        "...180"
## [181] "...181"                "Miami, FL"
## [183] "...183"                "...184"
## [185] "Spokane, WA"           "...186"
## [187] "...187"                "Salt Lake City, UT"
## [189] "...189"                "...190"
## [191] "Minneapolis, MN"       "...192"
## [193] "...193"                "Anaheim, CA"
## [195] "...195"                "...196"
## [197] "Providence, RI"        "...198"
## [199] "...199"                "Syracuse, NY"
## [201] "...201"                "...202"
## [203] "Peoria, AZ"            "...204"
## [205] "...205"                "Houston, TX"
## [207] "...207"                "...208"
## [209] "Prince George's County, MD" "...210"
## [211] "...211"
```

```
head(vice)
```

```
## # A tibble: 6 x 211
##   ...1      Denver ...3 ...4 'El Paso' ...6 ...7 Portland ...9
##   <dtm>      <chr>   <chr> <chr> <chr>   <chr> <chr> <chr>   <chr>
## 1 NA      Kia/Hyun~ All   Perc~ Kia/Hyun~ All   Perc~ Kia/Hyu~ All
## 2 2019-12-01 00:00:00 48.0    615.0 0.07~ 13.0    103.0 0.12~ 13.0    592.0
## 3 2020-01-01 00:00:00 21.0    519.0 0.04~ 9.0     95.0  0.09~ 12.0    559.0
## 4 2020-02-01 00:00:00 28.0    402.0 0.06~ 5.0     64.0  0.07~ 10.0    498.0
## 5 2020-03-01 00:00:00 35.0    508.0 0.06~ 6.0     75.0  0.08  4.0     465.0
## 6 2020-04-01 00:00:00 39.0    616.0 0.06~ 1.0     57.0  0.01~ 7.0     488.0
## # i 202 more variables: ...10 <chr>, Atlanta <chr>, ...12 <chr>, ...13 <chr>,
## #   Chicago <chr>, ...15 <chr>, ...16 <chr>, 'Virginia Beach' <chr>,
## #   ...18 <chr>, ...19 <chr>, Louisville <chr>, ...21 <chr>, ...22 <chr>,"
```

```
## # 'San Diego' <chr>, ...24 <chr>, ...25 <chr>, Sacramento <chr>, ...27 <chr>,
## # ...28 <chr>, Dallas <chr>, ...30 <chr>, ...31 <chr>, 'Fort Worth' <chr>,
## # ...33 <chr>, ...34 <chr>, Bakersfield <chr>, ...36 <chr>, ...37 <chr>,
## # 'Omaha, NE' <chr>, ...39 <chr>, ...40 <chr>, 'Lubbock, TX' <chr>, ...
```

```
str(vice)
```

```
## tibble [46 x 211] (S3: tbl_df/tbl/data.frame)
## $ ...1 : POSIXct[1:46], format: NA "2019-12-01" ...
## $ Denver : chr [1:46] "Kia/Hyundais" "48.0" "21.0" "28.0" ...
## $ ...3 : chr [1:46] "All" "615.0" "519.0" "402.0" ...
## $ ...4 : chr [1:46] "Percent" "0.07804878049" "0.04046242775" "0.06965174129" ...
## $ El Paso : chr [1:46] "Kia/Hyundais" "13.0" "9.0" "5.0" ...
## $ ...6 : chr [1:46] "All" "103.0" "95.0" "64.0" ...
## $ ...7 : chr [1:46] "Percent" "0.1262135922" "0.09473684211" "0.078125" ...
## $ Portland : chr [1:46] "Kia/Hyundais" "13.0" "12.0" "10.0" ...
## $ ...9 : chr [1:46] "All" "592.0" "559.0" "498.0" ...
## $ ...10 : chr [1:46] "Percent" "0.02195945946" "0.02146690519" "0.02008032129" ...
## $ Atlanta : chr [1:46] "Kia/Hyundais" NA NA NA ...
## $ ...12 : chr [1:46] "All" NA NA NA ...
## $ ...13 : chr [1:46] "Percent" NA NA NA ...
## $ Chicago : chr [1:46] "Kia/Hyundais" "46.0" "36.0" "40.0" ...
## $ ...15 : chr [1:46] "All" "767.0" "669.0" "653.0" ...
## $ ...16 : chr [1:46] "Percent" "0.05997392438" "0.05381165919" "0.06125574273" ...
## $ Virginia Beach : chr [1:46] "Kia/Hyundais" NA NA NA ...
## $ ...18 : chr [1:46] "All" NA NA NA ...
## $ ...19 : chr [1:46] "Percent" NA NA NA ...
## $ Louisville : chr [1:46] "Kia/Hyundais" "14.0" "18.0" "10.0" ...
## $ ...21 : chr [1:46] "All" NA "301.0" "277.0" ...
## $ ...22 : chr [1:46] "Percent" NA "0.05980066445" "0.03610108303" ...
## $ San Diego : chr [1:46] "Kia/Hyundais" "25.0" "14.0" "16.0" ...
## $ ...24 : chr [1:46] "All" "430.0" "423.0" "411.0" ...
## $ ...25 : chr [1:46] "Percent" "0.05813953488" "0.03309692671" "0.03892944039" ...
## $ Sacramento : chr [1:46] "Kia/Hyundais" "2.0" "9.0" "4.0" ...
## $ ...27 : chr [1:46] "All" NA "192.0" "157.0" ...
## $ ...28 : chr [1:46] "Percent" NA "0.046875" "0.02547770701" ...
## $ Dallas : chr [1:46] "Kia/Hyundais" "21.0" "26.0" "23.0" ...
## $ ...30 : chr [1:46] "All" NA "1060.0" "874.0" ...
## $ ...31 : chr [1:46] "Percent" NA "0.02452830189" "0.02631578947" ...
## $ Fort Worth : chr [1:46] "Kia/Hyundais" "12.0" "13.0" "4.0" ...
## $ ...33 : chr [1:46] "All" "352.0" "340.0" "310.0" ...
## $ ...34 : chr [1:46] "Percent" "0.03409090909" "0.03823529412" "0.01290322581" ...
## $ Bakersfield : chr [1:46] "Kia/Hyundais" "9.0" "13.0" "12.0" ...
## $ ...36 : chr [1:46] "All" "360.0" "393.0" "323.0" ...
## $ ...37 : chr [1:46] "Percent" "0.025" "0.03307888041" "0.03715170279" ...
## $ Omaha, NE : chr [1:46] "Kia/Hyundais" "11.0" "19.0" "13.0" ...
## $ ...39 : chr [1:46] "All" "277.0" "291.0" "266.0" ...
## $ ...40 : chr [1:46] "Percent" "0.03971119134" "0.06529209622" "0.04887218045" ...
## $ Lubbock, TX : chr [1:46] "Kia/Hyundais" "6.0" "2.0" "3.0" ...
## $ ...42 : chr [1:46] "All" "130.0" "101.0" "100.0" ...
## $ ...43 : chr [1:46] "Percent" "0.04615384615" "0.0198019802" "0.03" ...
## $ Fremont, CA : chr [1:46] "Kia/Hyundais" NA NA NA ...
## $ ...45 : chr [1:46] "All" "39.0" "63.0" "43.0" ...
## $ ...46 : chr [1:46] "Percent" NA NA NA ...
```

```

## $ San Antonio      : chr [1:46] "Kia/Hyundais" NA NA NA ...
## $ ...48            : chr [1:46] "All" NA NA NA ...
## $ ...49            : chr [1:46] "Percent" NA NA NA ...
## $ Irving, Texas    : chr [1:46] "Kia/Hyundais" "2.0" "3" "7" ...
## $ ...51            : chr [1:46] "All" "58.0" "78" "71" ...
## $ ...52            : chr [1:46] "Percent" "0.03448275862" "0.03846153846" "0.0985915493" ...
## $ Madison, WI      : chr [1:46] "Kia/Hyundais" "4.0" "4.0" "1.0" ...
## $ ...54            : chr [1:46] "All" "83.0" "75.0" "45.0" ...
## $ ...55            : chr [1:46] "Percent" "0.04819277108" "0.05333333333" "0.02222222222" ...
## $ Frisco, TX       : chr [1:46] "Kia/Hyundais" NA NA NA ...
## $ ...57            : chr [1:46] "All" "12.0" "15.0" "8.0" ...
## $ ...58            : chr [1:46] "Percent" NA NA NA ...
## $ Lincoln, NE      : chr [1:46] "Kia/Hyundais" NA NA NA ...
## $ ...60            : chr [1:46] "All" "28.0" "62.0" "51.0" ...
## $ ...61            : chr [1:46] "Percent" NA NA NA ...
## $ Wichita, KS      : chr [1:46] "Kia/Hyundais" "3.0" "4.0" "5.0" ...
## $ ...63            : chr [1:46] "All" "182.0" "167.0" "179.0" ...
## $ ...64            : chr [1:46] "Percent" "0.01648351648" "0.02395209581" "0.02793296089" ...
## $ Plano, TX        : chr [1:46] "Kia/Hyundais" "3.0" "1.0" "1.0" ...
## $ ...66            : chr [1:46] "All" "64.0" "42.0" "25.0" ...
## $ ...67            : chr [1:46] "Percent" "0.046875" "0.02380952381" "0.04" ...
## $ Akron, OH        : chr [1:46] "Kia/Hyundais" "9.0" "1.0" "2.0" ...
## $ ...69            : chr [1:46] "All" "67.0" "62.0" "42.0" ...
## $ ...70            : chr [1:46] "Percent" "0.1343283582" "0.01612903226" "0.04761904762" ...
## $ Buffalo, NY      : chr [1:46] "Kia/Hyundais" "7.0" "7.0" "9.0" ...
## $ ...72            : chr [1:46] "All" "107.0" "136.0" "100.0" ...
## $ ...73            : chr [1:46] "Percent" "0.06542056075" "0.05147058824" "0.09" ...
## $ Chula Vista, CA  : chr [1:46] "Kia/Hyundais" "6.0" "3.0" "2.0" ...
## $ ...75            : chr [1:46] "All" "77.0" "82.0" "50.0" ...
## $ ...76            : chr [1:46] "Percent" "0.07792207792" "0.03658536585" "0.04" ...
## $ Reno, NV         : chr [1:46] "Kia/Hyundais" NA NA NA ...
## $ ...78            : chr [1:46] "All" "82.0" "99.0" "68.0" ...
## $ ...79            : chr [1:46] "Percent" NA NA NA ...
## $ Oxnard, CA       : chr [1:46] "Kia/Hyundais" "1.0" "2.0" "0.0" ...
## $ ...81            : chr [1:46] "All" "48.0" "45.0" "28.0" ...
## $ ...82            : chr [1:46] "Percent" "0.02083333333" "0.04444444444" "0" ...
## $ San Francisco    : chr [1:46] "Kia/Hyundais" "11.0" "10.0" "9.0" ...
## $ ...84            : chr [1:46] "All" "312.0" "270.0" "285.0" ...
## $ ...85            : chr [1:46] "Percent" "0.03525641026" "0.03703703704" "0.03157894737" ...
## $ McKinney, TX     : chr [1:46] "Kia/Hyundais" "0.0" "0.0" "0.0" ...
## $ ...87            : chr [1:46] "All" "17.0" "16.0" "15.0" ...
## $ ...88            : chr [1:46] "Percent" "0" "0" "0" ...
## $ Arlington, TX    : chr [1:46] "Kia/Hyundais" "8.0" "3.0" "6.0" ...
## $ ...90            : chr [1:46] "All" "101.0" "87.0" "75.0" ...
## $ ...91            : chr [1:46] "Percent" "0.07920792079" "0.03448275862" "0.08" ...
## $ Garland, TX      : chr [1:46] "Kia/Hyundais" "3.0" "0.0" "0.0" ...
## $ ...93            : chr [1:46] "All" "112.0" "111.0" "78.0" ...
## $ ...94            : chr [1:46] "Percent" "0.02678571429" "0" "0" ...
## $ Riverside County, CA : chr [1:46] "Kia/Hyundais" "11.0" "10.0" "10.0" ...
## $ ...96            : chr [1:46] "All" "594.0" "583.0" "610.0" ...
## $ ...97            : chr [1:46] "Percent" "0.01851851852" "0.01715265866" "0.01639344262" ...
## $ Stockton, CA     : chr [1:46] "Kia/Hyundais" "3.0" "5.0" "3.0" ...
## $ ...99            : chr [1:46] "All" "119.0" "167.0" "143.0" ...
## [list output truncated]

```

```

#city with most thefts
colnames(vice)[1] <- "Date"
vice_data <- vice[!is.na(vice$Date), ]
#total for city
totals <- c()

for (i in seq(2, ncol(vice_data) - 2, by = 3)) {
  city_name <- colnames(vice_data)[i]
  all_col <- i + 1
  all_vals <- suppressWarnings(as.numeric(vice_data[[all_col]]))
  totals[city_name] <- sum(all_vals, na.rm = TRUE)
}

totals_sorted <- sort(totals, decreasing = TRUE)
head(totals_sorted, 5)

```

```

## Los Angeles, CA      Chicago      Dallas      Denver      Portland
##      86154      62226      48235      43014      32071

```

```

colnames(vice)[1] <- "Date"
vice_data <- vice[!is.na(vice$Date), ]

kia_totals <- c()

for (i in seq(2, ncol(vice_data) - 2, by = 3)) {
  city_name <- colnames(vice_data)[i]
  kia_vals <- suppressWarnings(as.numeric(vice_data[[i]]))
  kia_totals[city_name] <- sum(kia_vals, na.rm = TRUE)
}

kia_sorted <- sort(kia_totals, decreasing = TRUE)

head(kia_sorted, 5)

```

```

##      Chicago      Milwaukee, WI      Los Angeles, CA      Denver      Minneapolis, MN
##      14654      14403      13654      9925      6723

```

```

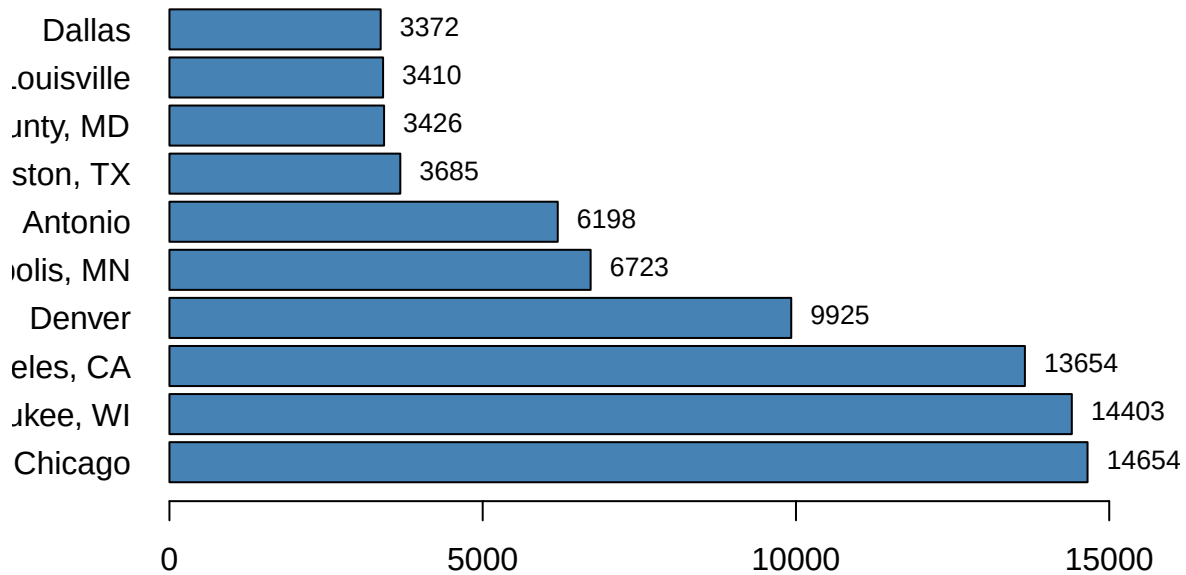
top_cities <- head(kia_sorted, 10)

bar_pos <- barplot(top_cities,
  horiz = TRUE,
  las = 1,
  col = "steelblue",
  main = "Top 10 Cities with Kia/Hyundai Thefts",
  xlab = "Total Number of Kia/Hyundai Thefts",
  xlim = c(0, max(top_cities) * 1.1))

text(x = top_cities,
  y = bar_pos,
  labels = round(top_cities),
  pos = 4,
  cex = 0.8,
  col = "black")

```

Top 10 Cities with Kia/Hyundai Thefts



Total Number of Kia/Hyundai Thefts

```
#stacked chart of top 5 cities with thefts
colnames(vice)[1] <- "Date"
vice_data <- vice[!is.na(vice$Date), ]

long <- data.frame(Date=as.Date(character()),
                  City=character(),
                  Kia=numeric(),
                  stringsAsFactors = FALSE)

for (i in seq(2, ncol(vice_data) - 2, by = 3)) {
  city_name <- colnames(vice_data)[i]
  kia_vals <- suppressWarnings(as.numeric(vice_data[[i]]))
  tmp <- data.frame(Date = as.Date(vice_data$Date),
                  City = city_name,
                  Kia = kia_vals,
                  stringsAsFactors = FALSE)
  long <- rbind(long, tmp)
}

long$Year <- as.integer(format(long$Date, "%Y"))
long <- long[long$Year >= 2019 & long$Year <= 2023, ]

yr_city <- aggregate(Kia ~ City + Year, data = long, sum, na.rm = TRUE)

tot_city <- aggregate(Kia ~ City, data = yr_city, sum)
top5 <- head(tot_city[order(-tot_city$Kia), "City"], 5)
```

```

yr_top <- yr_city[yr_city$City %in% top5, ]

tab <- with(yr_top, tapply(Kia, list(as.character(Year), City), sum, na.rm = TRUE))
tab[is.na(tab)] <- 0

yrs <- as.character(2019:2023)
tab <- tab[intersect(yrs, rownames(tab)), top5, drop = FALSE]

rownames(tab) <- as.character(sort(as.integer(rownames(tab))))
year_colors <- c("2019" = "#4575b4",
                 "2020" = "#74add1",
                 "2021" = "#fdae61",
                 "2022" = "#f46d43",
                 "2023" = "#d73027")

bar_colors <- year_colors[rownames(tab)]

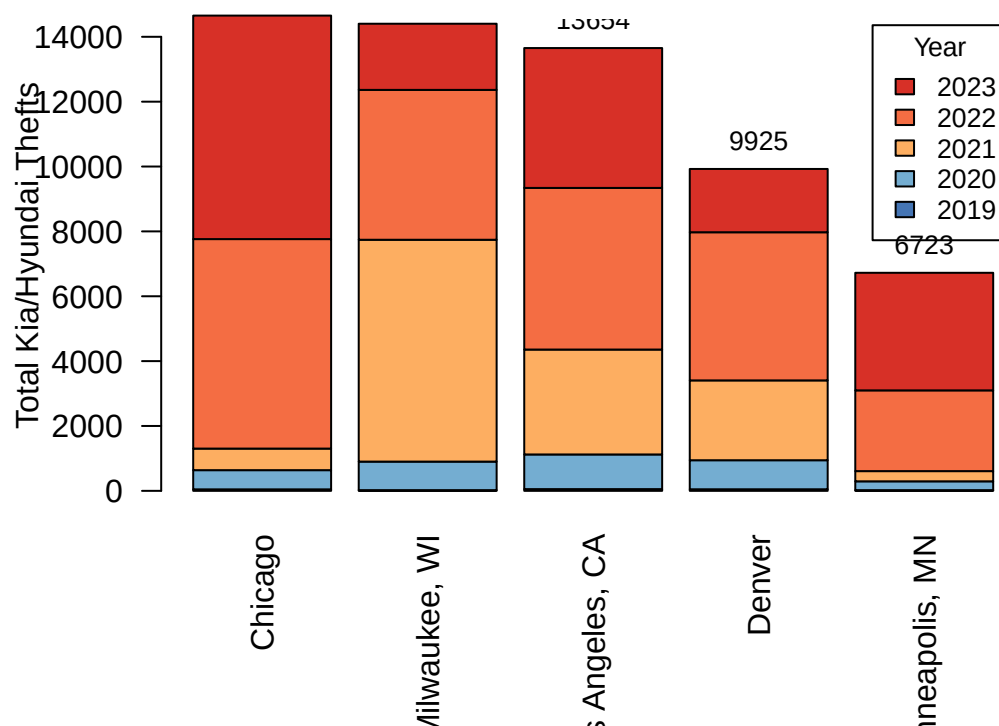
op <- par(mar = c(6, 8, 4, 2))
bp <- barplot(tab,
              beside = FALSE,
              col = bar_colors,
              legend.text = rownames(tab),
              args.legend = list(x = "topright", inset = 0.02, cex = 0.8, title = "Year"),
              main = "Kia/Hyundai Thefts by Year: Top 5 Cities",
              ylab = "Total Kia/Hyundai Thefts",
              las = 2)

par(op)

city_totals <- colSums(tab)
text(x = bp, y = city_totals, labels = city_totals, pos = 3, cex = 0.8)

```


Kia/Hyundai Thefts by Year: Top 5 Cities



```
#tree map
library(treemap)

colnames(vice)[1] <- "Date"
vice_data <- vice[!is.na(vice$Date), ]

long <- data.frame(Date = as.Date(character()),
                  City = character(),
                  Kia = numeric(),
                  stringsAsFactors = FALSE)

for (i in seq(2, ncol(vice_data) - 2, by = 3)) {
  city_name <- colnames(vice_data)[i]
  kia_vals <- suppressWarnings(as.numeric(vice_data[[i]]))
  tmp <- data.frame(Date = as.Date(vice_data$Date),
                  City = city_name,
                  Kia = kia_vals)
  long <- rbind(long, tmp)
}

long$Year <- as.integer(format(long$Date, "%Y"))

long <- long[!is.na(long$Kia) & long$Year >= 2019 & long$Year <= 2023, ]

long$Kia <- as.numeric(long$Kia)
```

```
city_year <- aggregate(Kia ~ City + Year, data = long, sum, na.rm = TRUE)
```

```
totals <- aggregate(Kia ~ City, data = city_year, sum)
```

```
top10 <- head(totals[order(-totals$Kia), "City"], 10)
```

```
tree_data <- subset(city_year, City %in% top10)
```

```
print(head(tree_data))
```

```
##           City Year Kia
## 11      Chicago 2019  46
## 14      Dallas 2019  21
## 15      Denver 2019  48
## 22  Houston, TX 2019  24
## 24 Los Angeles, CA 2019 53
## 25    Louisville 2019  14
```

```
tree_plot <- subset(tree_data, !is.na(Kia) & Kia > 0)
```

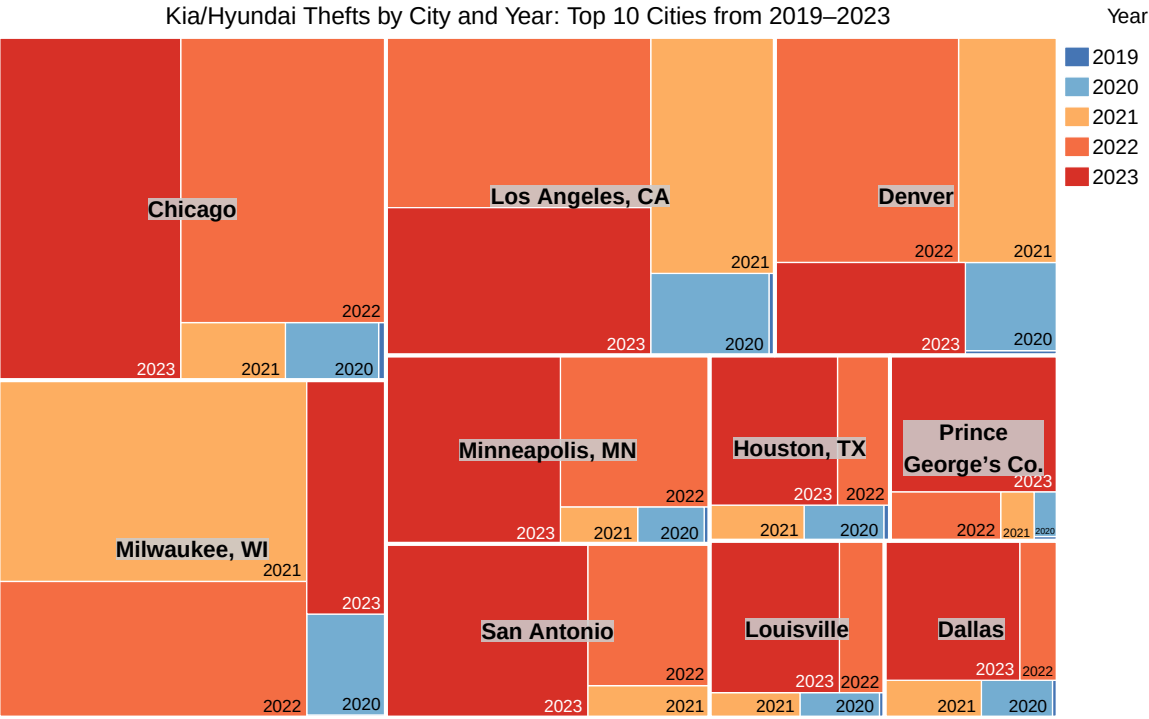
```
tree_plot$Year <- factor(tree_plot$Year, levels = 2019:2023)
```

```
tree_plot$City <- gsub("Prince George's County, MD", "Prince George's Co.", tree_plot$City)
```

```
year_colors <- c("2019" = "#4575b4",
                 "2020" = "#74add1",
                 "2021" = "#fdae61",
                 "2022" = "#f46d43",
                 "2023" = "#d73027")
```

```
treemap(tree_plot,
  index = c("City", "Year"),
  vSize = "Kia",
  vColor = "Year",
  type = "categorical",
  palette = unname(year_colors[levels(tree_plot$Year)]),
  title = "Kia/Hyundai Thefts by City and Year: Top 10 Cities from 2019-2023",
  fontsize.labels = c(13, 10),
  align.labels = list(c("center","center"), c("right","bottom")),
  border.col = "white")
```

Kia/Hyundai Thefts by City and Year: Top 10 Cities from 2019–2023



week5&6_part2

Joanna Sierra-Mendoza

2025-10-11

```
library(readxl)
df_map <- read.csv("carTheftsMap.csv")
head(df_map, 10)
```

```
##      agency_ori      geo_name countCarThefts2019
## 1    M00490300      Carthage PD                62
## 2              Warren County SO                112
## 3      TX06802      Odessa PD                499
## 4    M00530000      Laclede County SO             55
## 5    M00480000      Jackson County SO            125
## 6    MN0070100 Mankato Dept of Public Safety       72
## 7              Pickaway County SO             50
## 8              Roy PD              60
## 9    RI0040400      East Providence PD           73
## 10   MN0860000      Wright County SO            65
##      countCarThefts2020 countCarThefts2021 countCarThefts2022 latitude longitude
## 1              58              47              26 37.17585 -94.31335
## 2              94              76              58 41.30879 -80.84788
## 3             464             375             288 31.88933 -102.32989
## 4              74              54              32 37.66346 -92.54901
## 5             141              81              75 37.38440 -89.66792
## 6              62              66              44 44.16168 -94.00876
## 7              37              35              31 39.65586 -83.04868
## 8              61              60              37 41.17156 -112.02728
## 9              42              53              46 41.84003 -71.46641
## 10             65              43              42 45.20876 -93.86329
##      percentChange2019to2022
## 1             -0.5806452
## 2             -0.4821429
## 3             -0.4228457
## 4             -0.4181818
## 5             -0.4000000
## 6             -0.3888889
## 7             -0.3800000
## 8             -0.3833333
## 9             -0.3698630
## 10            -0.3538462
```

```
names(df_map)
```

```
## [1] "agency_ori"      "geo_name"
```

```
## [3] "countCarThefts2019"      "countCarThefts2020"
## [5] "countCarThefts2021"      "countCarThefts2022"
## [7] "latitude"                "longitude"
## [9] "percentChange2019to2022"
```

```
library(dplyr)
library(tidyr)

df_map_clean <- df_map %>%
  mutate(across(starts_with("countCarThefts"),
    ~ as.numeric(gsub(",", "", as.character(.)))))

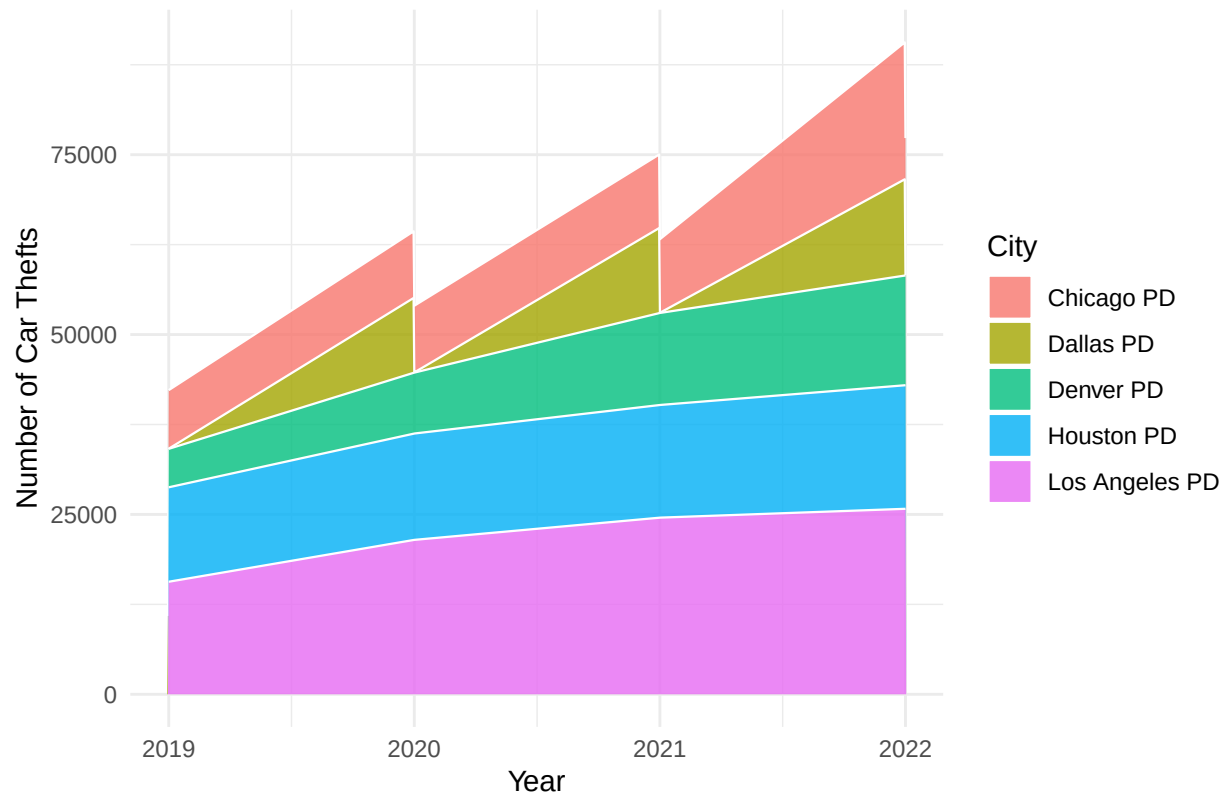
df_long <- df_map_clean %>%
  select(geo_name, starts_with("countCarThefts")) %>%
  pivot_longer(
    cols = starts_with("countCarThefts"),
    names_to = "Year",
    values_to = "Thefts"
  ) %>%
  mutate(
    Year = as.integer(sub("countCarThefts", "", Year)),
    Thefts = as.numeric(Thefts)
  )

top5 <- df_long %>%
  group_by(geo_name) %>%
  summarise(total = sum(Thefts, na.rm = TRUE), .groups = "drop") %>%
  slice_max(total, n = 5) %>%
  pull(geo_name)

df_top5 <- df_long %>% filter(geo_name %in% top5)

library(ggplot2)
ggplot(df_top5, aes(x = Year, y = Thefts, fill = geo_name)) +
  geom_area(alpha = 0.8, color = "white", linewidth = 0.4) +
  scale_x_continuous(breaks = 2019:2022) +
  labs(
    title = "Car Thefts by Year: Top 5 Cities",
    x = "Year", y = "Number of Car Thefts",
    fill = "City"
  ) +
  theme_minimal()
```

Car Thefts by Year: Top 5 Cities



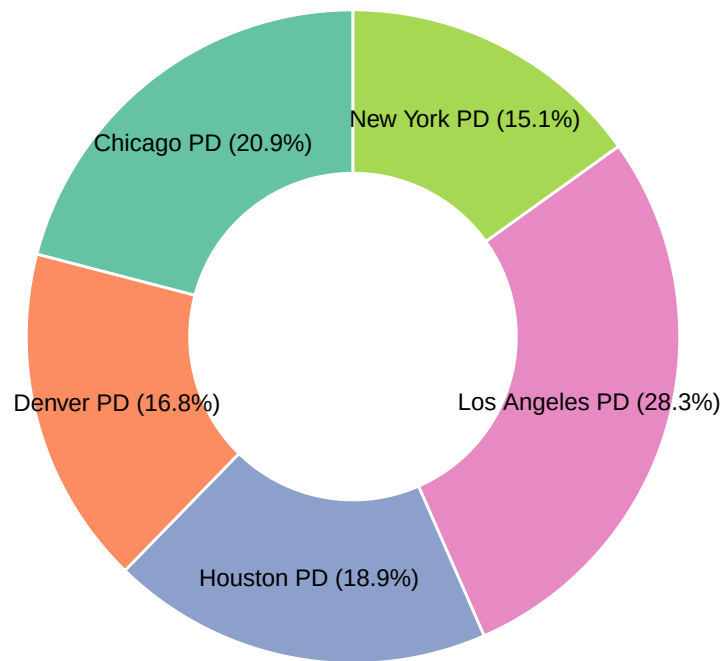
```
#pie chart
df_2022 <- df_map %>%
  select(geo_name, countCarThefts2022) %>%
  mutate(countCarThefts2022 = as.numeric(gsub(",", "", as.character(countCarThefts2022))))

top5_2022 <- df_2022 %>%
  arrange(desc(countCarThefts2022)) %>%
  slice_head(n = 5)

top5_2022 <- top5_2022 %>%
  mutate(percentage = round(100 * countCarThefts2022 / sum(countCarThefts2022), 1),
         label = paste0(geo_name, " (", percentage, "%)"))

ggplot(top5_2022, aes(x = 2, y = countCarThefts2022, fill = geo_name)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar(theta = "y", start = 0) +
  xlim(0.5, 2.5) +
  theme_void() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            size = 3) +
  labs(title = "Top 5 Cities for Car Thefts in 2022") +
  scale_fill_brewer(palette = "Set2") +
  theme(legend.position = "none")
```

Top 5 Cities for Car Thefts in 2022

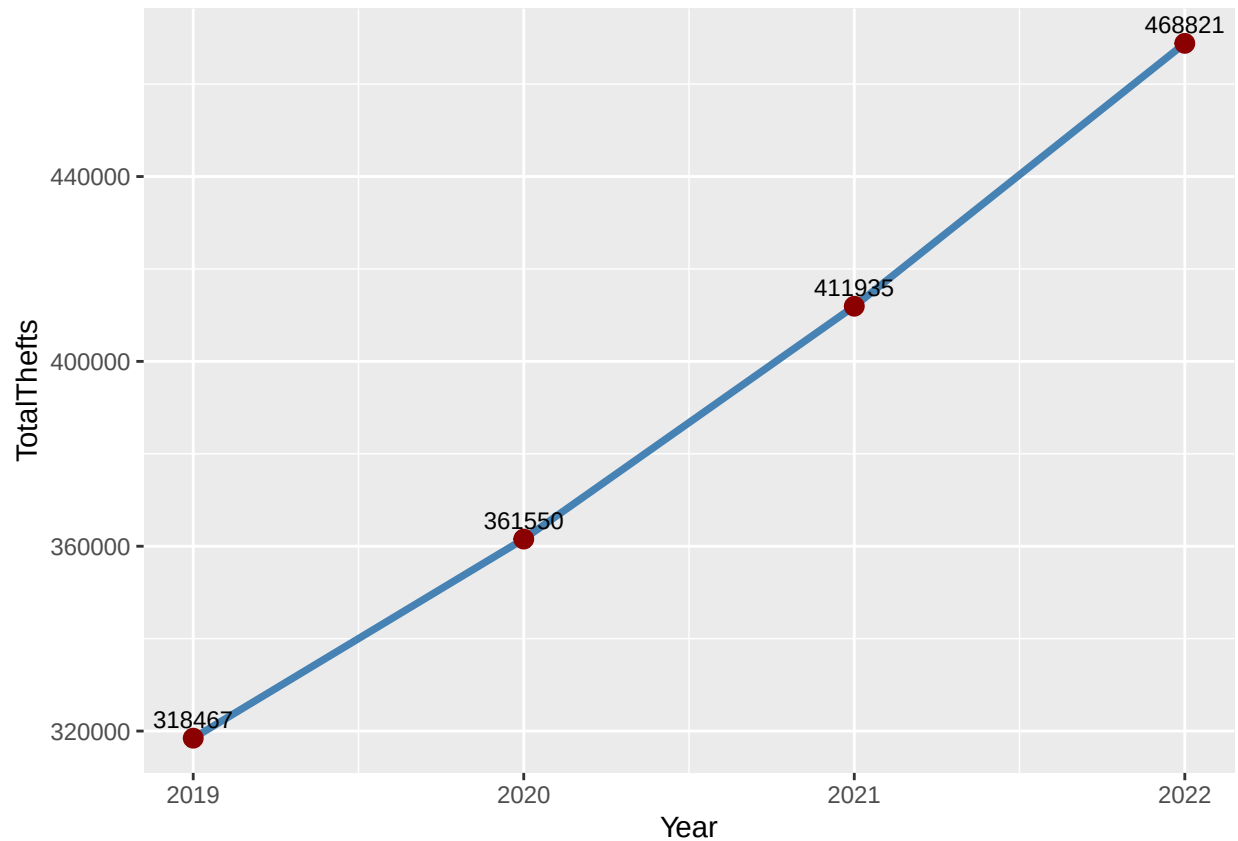


```
#line chart
df_map_clean <- df_map %>%
  mutate(across(starts_with("countCarThefts"),
    ~ as.numeric(gsub(",", "", as.character(.)))))

year_totals <- df_map_clean %>%
  summarise(
    `2019` = sum(countCarThefts2019, na.rm = TRUE),
    `2020` = sum(countCarThefts2020, na.rm = TRUE),
    `2021` = sum(countCarThefts2021, na.rm = TRUE),
    `2022` = sum(countCarThefts2022, na.rm = TRUE)
  )

year_totals_long <- year_totals %>%
  tidyr::pivot_longer(cols = everything(),
    names_to = "Year",
    values_to = "TotalThefts") %>%
  mutate(Year = as.integer(Year))

ggplot(year_totals_long, aes(x = Year, y = TotalThefts)) +
  geom_line(color = "steelblue", size = 1.3) +
  geom_point(size = 3, color = "darkred") +
  geom_text(aes(label = TotalThefts), vjust = -0.6, size = 3)
```



```
labs(title = "Total Car Thefts Reported by all Police Departments",
      x = "Year",
      y = "Total Number of Thefts") +
theme_minimal() +
scale_x_continuous(breaks = 2019:2022)
```

```
## NULL
```

```
#percent change geographical overview pie chart
df_map$percentChange2019to2022 <- as.numeric(gsub("%", "", as.character(df_map$percentChange2019to2022)))

df_map <- df_map %>%
  mutate(
    Region = case_when(
      latitude < 33 ~ "South",
      latitude >= 37 & longitude > -100 & longitude < -80 ~ "Midwest",
      longitude <= -100 ~ "West",
      longitude >= -80 ~ "East",
      TRUE ~ "North"
    ),
    Region = factor(Region, levels = c("North", "Midwest", "South", "East", "West"))
  )

region_summary <- df_map %>%
  group_by(Region) %>%
```



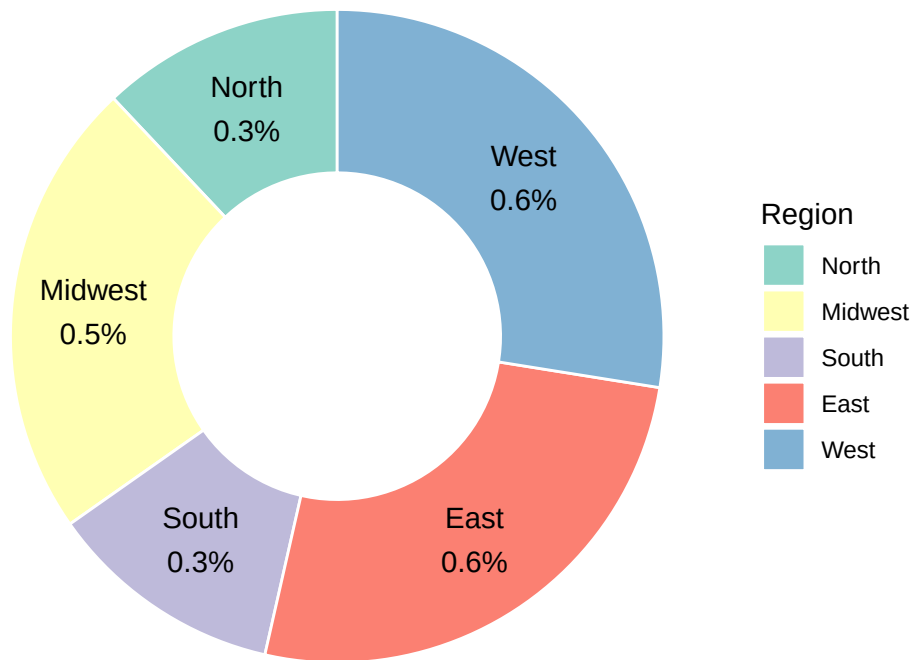
```

summarise(avg_percent_change = mean(percentChange2019to2022, na.rm = TRUE)) %>%
filter(!is.na(Region))

ggplot(region_summary, aes(x = 2, y = avg_percent_change, fill = Region)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar(theta = "y", start = 0) +
  xlim(0.5, 2.5) +
  theme_void() +
  geom_text(aes(label = paste0(Region, "\n", round(avg_percent_change, 1), "%")),
            position = position_stack(vjust = 0.5),
            size = 4) +
  labs(title = "Average % Change in Car Thefts (2019–2022) by U.S. Region") +
  scale_fill_brewer(palette = "Set3")

```

Average % Change in Car Thefts (2019–2022) by U.S. Region



```

#pie chart of total thefts per region

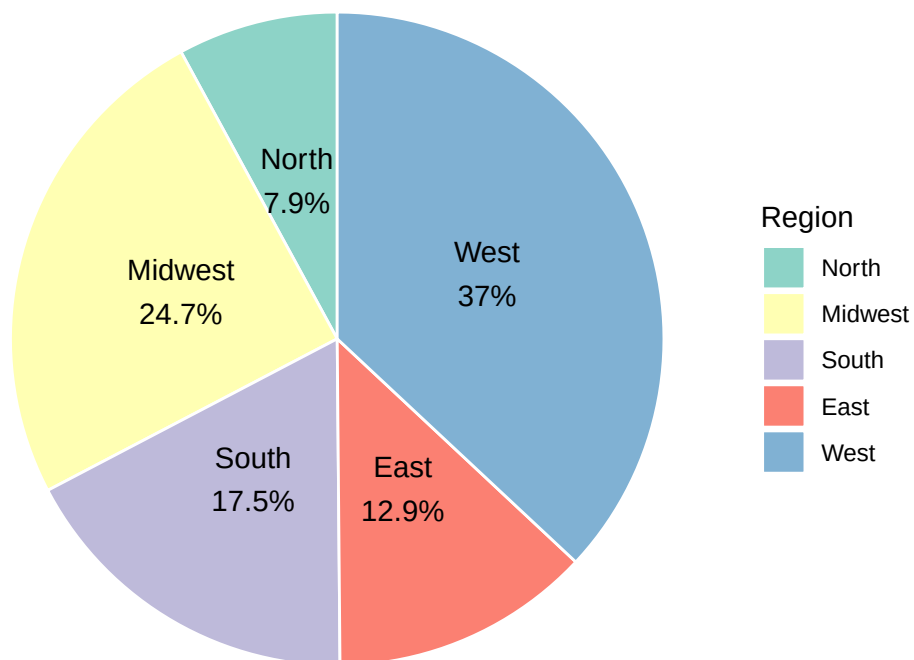
df_map_clean <- df_map %>%
  mutate(across(starts_with("countCarThefts"),
    ~ as.numeric(gsub(",", "", as.character(.)))))

region_totals <- df_map_clean %>%
  group_by(Region) %>%
  summarise(total_thefts = sum(countCarThefts2019 + countCarThefts2020 +
    countCarThefts2021 + countCarThefts2022, na.rm = TRUE)) %>%
  filter(!is.na(Region))

```

```
ggplot(region_totals, aes(x = "", y = total_thefts, fill = Region)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar(theta = "y") +
  theme_void() +
  geom_text(aes(label = paste0(Region, "\n", round(100 * total_thefts / sum(total_thefts), 1), "%")),
            position = position_stack(vjust = 0.5),
            size = 4) +
  labs(title = "Total Car Thefts (2019-2022) by Region") +
  scale_fill_brewer(palette = "Set3")
```

Total Car Thefts (2019–2022) by Region



```
library(readxl)
df_kia <- read.csv("kiaHyundaiThefts.csv")
head(df_kia, 10)
```

##	month	year	city	state	countKiaHyundaiThefts	countOtherThefts
## 1	Jan	2019	Atlanta	GA	17	264
## 2	Feb	2019	Atlanta	GA	11	205
## 3	Mar	2019	Atlanta	GA	18	181
## 4	Apr	2019	Atlanta	GA	15	223
## 5	May	2019	Atlanta	GA	16	277
## 6	Jun	2019	Atlanta	GA	14	220
## 7	Jul	2019	Atlanta	GA	19	267
## 8	Aug	2019	Atlanta	GA	12	242
## 9	Sep	2019	Atlanta	GA	11	234

```
## 10    Oct 2019 Atlanta    GA                12                206
##      percentKiaHyundai
## 1              0.060
## 2              0.051
## 3              0.090
## 4              0.063
## 5              0.055
## 6              0.060
## 7              0.066
## 8              0.047
## 9              0.045
## 10             0.055
```

```
names(df_kia)
```

```
## [1] "month"          "year"           "city"
## [4] "state"          "countKiaHyundaiThefts" "countOtherThefts"
## [7] "percentKiaHyundai"
```

```
#pie chart of Kia thefts
```

```
df_kia$countKiaHyundaiThefts <- as.numeric(gsub(",", "", as.character(df_kia$countKiaHyundaiThefts)))
```

```
to_abbr <- function(x) {
  x <- toupper(trimws(x))
  ab_ok <- x %in% state.abb
  if (all(ab_ok | x %in% c("DC"))) return(x)
  out <- x
  map <- setNames(state.abb, toupper(state.name))
  out[!ab_ok] <- ifelse(!is.na(map[out[!ab_ok]]), map[out[!ab_ok]], out[!ab_ok])
  out[out == "DISTRICT OF COLUMBIA"] <- "DC"
  out
}
```

```
df_kia$state_abbr <- to_abbr(df_kia$state)
```

```
north_states <- c("ME", "NH", "VT", "MA", "RI", "CT", "NY")
east_states <- c("NJ", "PA", "DE", "MD", "DC", "VA", "WV")
midwest_states <- c("OH", "MI", "IN", "IL", "WI", "MN", "IA", "MO", "ND", "SD", "NE", "KS")
south_states <- c("KY", "TN", "NC", "SC", "GA", "FL", "AL", "MS", "LA", "AR", "OK", "TX")
west_states <- c("CA", "OR", "WA", "NV", "AZ", "NM", "CO", "UT", "ID", "MT", "WY", "AK", "HI")
```

```
df_kia <- df_kia %>%
  mutate(Region = case_when(
    state_abbr %in% north_states ~ "North",
    state_abbr %in% east_states ~ "East",
    state_abbr %in% midwest_states ~ "Midwest",
    state_abbr %in% south_states ~ "South",
    state_abbr %in% west_states ~ "West",
    TRUE ~ "Other"
  ))
```

```
region_totals_kia <- df_kia %>%
  group_by(Region) %>%
  summarise(total_thefts = sum(countKiaHyundaiThefts, na.rm = TRUE), .groups = "drop") %>%
```

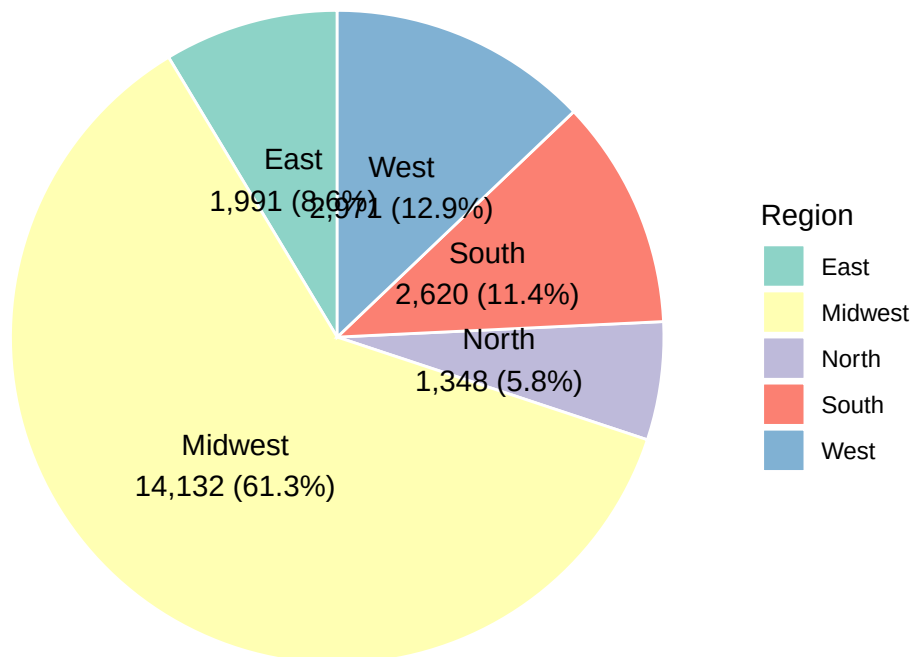
```

filter(Region %in% c("North", "East", "Midwest", "South", "West")) %>%
  arrange(match(Region, c("North", "East", "Midwest", "South", "West")))

ggplot(region_totals_kia, aes(x = "", y = total_thefts, fill = Region)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar(theta = "y") +
  theme_void() +
  geom_text(aes(label = paste0(Region, "\n", scales::comma(total_thefts),
                              " (", round(100 * total_thefts / sum(total_thefts), 1), "%)")),
            position = position_stack(vjust = 0.5), size = 3.8) +
  labs(title = "Kia/Hyundai Car Thefts (2019–2022) by Region") +
  scale_fill_brewer(palette = "Set3")

```

Kia/Hyundai Car Thefts (2019–2022) by Region



```

library(readxl)
df_mil <- read.csv("KiaHyundaiMilwaukeeData.csv")
head(df_mil, 10)

```

##	month	year	city	state	countKiaHyundaiThefts	countOtherThefts
## 1	Jan	2019	Milwaukee	WI	22	235
## 2	Feb	2019	Milwaukee	WI	13	218
## 3	Mar	2019	Milwaukee	WI	10	195
## 4	Apr	2019	Milwaukee	WI	10	238
## 5	May	2019	Milwaukee	WI	11	280
## 6	Jun	2019	Milwaukee	WI	15	330
## 7	Jul	2019	Milwaukee	WI	25	295

```
## 8      Aug 2019 Milwaukee      WI                21                352
## 9      Sep 2019 Milwaukee      WI                18                256
## 10     Oct 2019 Milwaukee      WI                25                266
##      percentKiaHyundai
## 1              0.086
## 2              0.056
## 3              0.049
## 4              0.040
## 5              0.038
## 6              0.043
## 7              0.078
## 8              0.056
## 9              0.066
## 10             0.086
```

```
names(df_mil)
```

```
## [1] "month"          "year"           "city"
## [4] "state"          "countKiaHyundaiThefts" "countOtherThefts"
## [7] "percentKiaHyundai"
```

```
df_mil <- df_mil %>%
  mutate(
    month_num = match(tolower(substr(month, 1, 3)), tolower(month.abb)),
    date = as.Date(paste(year, month_num, "01", sep = "-"))
  )

ggplot(df_mil, aes(x = date)) +
  geom_line(aes(y = countKiaHyundaiThefts, color = "Kia/Hyundai Thefts"), size = 1.2) +
  geom_line(aes(y = countOtherThefts, color = "Other Vehicle Thefts"), size = 1.2) +
  labs(
    title = "Kia/Hyundai vs. Other Vehicle Thefts in Milwaukee",
    x = "Month-Year",
    y = "Number of Thefts",
    color = "Theft Type"
  ) +
  scale_x_date(date_labels = "%b %Y", date_breaks = "6 months") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

